`# ShiftSafeguard — Parametric Income Insurance for E-Commerce Delivery Partners

Guidewire DEVTrails 2026 | Submission

⚡ The Evolution: From GigShield to ShiftSafeguard

> [!IMPORTANT] **Why we rebranded**: While *GigShield* was built to "shield" individual tasks, we realized that the true problem isn't just a missed delivery—it's the lost **shift**.
>
> **ShiftSafeguard** reflects our broader, more proactive mission: safeguarding the entire livelihood of workers by protecting their most valuable asset—**time**. We don't just shield the gig; we safeguard the shift.

🏗️ System Architecture

The ShiftSafeguard engine is built for scale, balancing instant payouts with a layered adversarial defense. Here's how the "Arc" works:

1. **Telemetry Ingestion**: Grabs a full stack of signal data (GPS, Sensor, Network).
2. **Stage 1: Fast Path**: Redis-backed static rule engine filters known spoofers/emulators in milliseconds.
3. **Stage 2: ML Pipeline & Voting**: Our democratic ensemble (Vision + NLP) adjudicates the core claim.
4. **Green/Yellow Route**: Verified claims hit UPI instantly; suspicious ones trigger an in-app Step-Up Challenge (real-time video).

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1. The Problem

Amazon Flex and Flipkart delivery partners are the last mile of India's infrastructure. They operate in unpredictable outdoor environments — extreme heat, heavy rain, floods, dense smog, curfews — with **no income protection**.

A Zomato worker has platform visibility; an Amazon Flex partner has nothing. When a red-alert day wipes out their shift, they absorb 100% of the loss.

ShiftSafeguard insures the income lost during these events. Nothing else.

- **Coverage scope**: LOSS OF INCOME ONLY.
- **Explicitly excluded**: Vehicle repairs, health insurance, accident medical bills.

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2. Persona & Scenario Walkthrough

Persona: Ravi, 27. Amazon Flex delivery partner, Pune.
Earns ₹600–800/day. Supports a family of 4.

Scenario A — Extreme Heat Day

IMD issues a heat alert: temp > 43°C in Ravi's zone. Amazon Flex suspends outdoor deliveries. Ravi loses a full day's earnings.

- **ShiftSafeguard Response**: Weather API detects threshold breach; policy activates automatically (no claim needed); ₹500 payout hits his UPI in < 2 minutes.

Scenario B — Flash Flood / Red Alert Rain

Heavy rainfall causes waterlogging. Deliveries suspended for 6 hours.

- **ShiftSafeguard Response**: Rain API confirms disruption; proportional payout issued; fraud layer cross-checks Ravi's GPS zone.

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3. Weekly Premium Model

Gig workers operate week-to-week. Monthly models don’t fit their cash flow. ShiftSafeguard is priced and settled weekly.

Plan	Weekly Premium	Max Weekly Payout	Coverage
Basic Shield	₹29	₹500	1 disruption/week
Standard Shield	₹59	₹1,200	Up to 3 disruptions/week
Pro Shield	₹99	₹2,000	Unlimited disruptions/week

Dynamic Premium Calculation (AI Layer): Base premium is adjusted weekly using an **XGBoost** model:

- **Zone Risk Score**: Historical frequency in worker's delivery zone.
- **Seasonal Modifier**: Monsoon/Summer surcharge; Winter discount.
- **Tenure Discount**: Loyal workers with clean history get lower premiums.

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4. The Brain: Two-Stage AI Pipeline

Stage 1: Fast Path (Heuristic + Unsupervised ML)

Redis-backed static rules and **DBSCAN Clustering** check:

- **DBSCAN Anomaly Detection**: Clusters GPS pings to identify spatial-temporal "teleportation" or algorithmic spoofing.
- **OS Mock Location Flag**: Catches simple GPS spoofers.
- **Emulator Signatures**: Blocks dev-tool driven attacks.

Stage 2: Multimodal Ensemble (The "Democratic Vote")

Claims that pass Stage 1 enter our **Multimodal Ensemble AI Pipeline**:

- **Visual Stream**: Convolutional Vision model checks image integrity and pixel-level consistency.
- **Contextual NLP Stream**: Analyzes description semantic intent against live zone-disruption reports.
- **Probabilistic Fusion**: A 50/50 weighted vote resolve both streams into a confidence score (>85% for Green Route).
- **Decision Logic**: High-confidence claims trigger instant UPI settlement; marginal ones enter "Yellow Route" for human audit.

📈 Parametric Triggers

Triggers are automated, zone-specific, and requiring zero manual filing.

Trigger	Data Source	Threshold	Payout Type
Extreme Heat	OpenWeatherMap / IMD	> 44.0 °C	Full Day
Heavy Rain	OpenWeatherMap / IMD	> 15.0 mm/hr	Proportional
AQI (Pollution)	CPCB / OpenWeather	> 300 (Very Poor)	Full Day
Curfew / Strike	Admin Flagged	Manual Trigger	Full Day

🛠️ Tech Stack

- **Frontend**: Next.js 14, Tailwind CSS, Framer Motion.
- **Backend**: FastAPI (Python), Uvicorn.
- **ML Engine**: Python, XGBoost, Scikit-learn (DBSCAN + Behavioral Velocity).,OPENWEATHERMAP API

- **Payments**: Razorpay Mock UPI Integration.

🏗️ Detailed Repository Structure

```
ShiftSafeguard/
├── frontend/                     # Next.js 14 — worker app + admin dashboard
│   ├── app/
│   │   ├── (worker)/             # Worker-facing flows
│   │   │   ├── dashboard/        # Coverage & payouts
│   │   │   ├── onboarding/       # Registration & Aadhaar KYC
│   │   │   └── policy/           # Weekly plan selection
│   │   ├── (admin)/              # Admin-facing analytics
│   │   │   ├── dashboard/        # Loss ratios & fraud flags
│   │   │   └── zones/            # Zone/Manual trigger control
│   │   ├── layout.tsx
│   │   └── page.tsx              # Entry landing
│   ├── components/               # Custom UI (PolicyCard, PayoutStatus, etc.)
│   └── lib/                      # API Clients & Utilities
├── backend/                      # FastAPI Microservice — The AI Brain
│   ├── main.py                   # Orchestrator & entry point
│   ├── src/
│   │   ├── api/                  # Routes (Premium, Claims, Triggers)
│   │   ├── services/             # Core Logic (Ensemble AI, Fraud, Trigger)
│   │   ├── schemas/              # Pydantic data models
│   │   └── models/               # AI Models (Ensemble .joblib & Pricing .pkl)
│   ├── requirements.txt
│   └── .env.example
├── docs/                         # Technical Specs, Diagrams & PDF Reports
│   ├── Blank diagram - Page 1.svg# Master Architecture Visual
│   ├── adversarial-defense.pdf   # GPS Spoofing Defense Spec
│   └── Original_Project_Spec.md  # Foundational Phase 1 Doc
├── docker-compose.yml            # Deployment orchestration
└── README.md                     # This file
```

👨‍💻 Team ShiftSafeguard

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ShiftSafeguard — Because the last mile shouldn't be the first to bear the loss. `